universe increased by at least a factor of 10^{30} during inflation, which means the volume of the universe increased by a factor of at least $(10^{30})^3 = 10^{90}$. Consequently, the energy embodied in the inflaton field *increased* by the same huge factor: as the inflationary phase drew to a close, a mere 10^{-35} or so seconds after it began, the energy in the inflaton field grew by a factor on the order of 10^{90}, if not more. This means that at the onset of inflation, the *inflaton field didn't need to have much energy, since the enormous expansionit was about to spawn would enormously amplify the energy it carried.* A simple calculation shows that a tiny nugget, on the order of 10^{-26} centimeters across, filled with a uniform inflaton field—and weighing a mere *twenty pounds*—would, through the ensuing inflationary expansion, acquire enough energy to account for all we see in the universe today.[2]

Thus, in stark contrast to the standard big bang theory in which the total mass/energy of the early universe was huge beyond words, inflationary cosmology, by "mining" gravity, can produce all the ordinary matter and radiation in the universe from a tiny, twenty-pound speck of inflatonfilled space. By no means does this answer Leibniz's question of why there is something rather than nothing, since we've yet to explain why there is an inflaton or even the space it occupies. But the something in need of explanation weighs a whole lot less than my dog Rocky, and that's certainly a very different starting point than envisaged in the standard big bang.[28]

Inflation, Smoothness, and the Arrow of Time

Perhaps my enthusiasm has already betrayed my bias, but of all the progress that science has achieved in our age, advances in cosmology fill me with the greatest awe and humility. I seem never to have lost the rush I initially felt years ago when I first read up on the basics of general relativity and realized that from our tiny little corner of spacetime we can apply Einstein's theory to learn about the evolution of the entire cosmos. Now, a few decades later, technological progress is subjecting these once abstract proposals for how the universe behaved in its earliest moments to observational tests, and the theories *really work.*